9B Lesson 5

# Growing crops

Year 9 Science · 9B Plant growth · Student Book pages 28 to 31

Name _____ Class _____ Date _____

**Today I can:**

- ☐ I can explain how fertilisers, pesticides, greenhouses and machines increase crop yield, and how pests and population growth change the food supply.
- ☐ I can describe how new varieties are made by cross-breeding and selective breeding, and interpret the results of a fertiliser investigation.

**Key words**

**yield** the amount of useful product a farmer gets from a crop
**fertiliser** a substance added to soil that contains mineral salts to help plants grow
**manure** animal waste used as a natural fertiliser
**decomposer** a microorganism that breaks down dead material and waste
**pesticide** a substance that kills pests (organisms that damage crops)
**insecticide** a pesticide that kills insects
**fungicide** a pesticide that kills fungi that cause plant diseases
**herbicide** a pesticide that kills weeds
**weed** a plant growing where it is not wanted, competing with the crop
**variety** a group of plants bred to have certain characteristics
**selective breeding** breeding only from the plants with the wanted characteristics, generation after generation

## ● Core Everyone does this section.

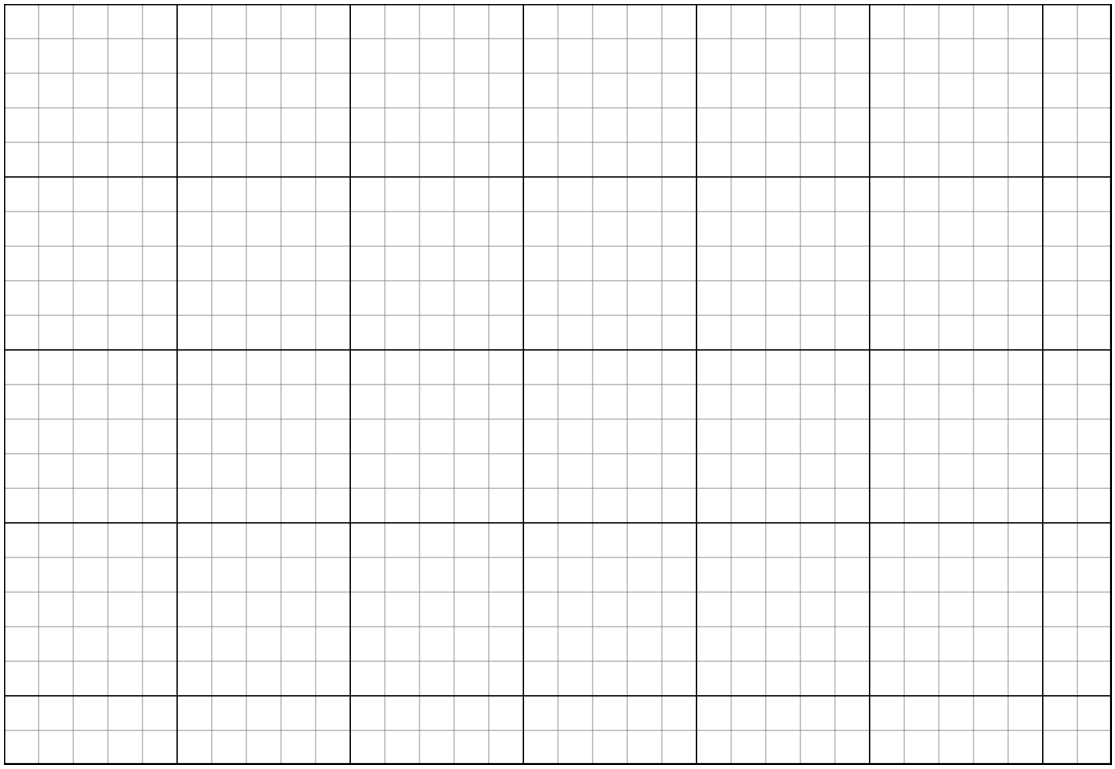
**1.** Match each type of pesticide to what it kills. Draw a line.

| | |
|---|---|
| insecticide | weeds with broad leaves, but not narrow-leaved crops such as wheat |
| fungicide | insect pests such as aphids |
| herbicide | all weeds (plants competing with the crop) |
| selective herbicide | fungi that cause plant diseases such as rust |

**2.** Students investigated fertiliser. Each group grew 20 bean plants in identical pots, one plant per pot, and added a different number of drops of the same liquid fertiliser. After a week they calculated the mean increase in height.

**a)** Plot the results as a bar chart on the grid. Label both axes.

| Group | Drops of liquid fertiliser | Mean increase in height (cm) |
|---|---|---|
| 1 | 2 | 4.6 |
| 2 | 4 | 6.3 |
| 3 | 6 | 7.8 |
| 4 | 8 | 7.2 |



**b)** Describe the trend in the results between groups 1, 2 and 3.

**c)** What advice would you give a farmer about how much fertiliser to use, based on the result for group 4? Give a reason.

**3.** Complete the sentences. Use the word bank.

Fertilisers contain _____ _____ that help plants grow. Farmers use cheap _____ fertilisers and natural ones such as _____ (animal waste).

Microorganisms called _____ break down manure and release the mineral salts, but this takes a long time.

## ●● Stretch When Core is finished.

**S1.** Aphids are insect pests. They suck the sugary liquid (sap) from the phloem of plants. Explain how spraying a crop to kill aphids could increase the crop yield.

**S2.** Selective herbicides kill weeds with broad leaves but not crops with narrow leaves.

**a)** Suggest why a wheat farmer might want to use a selective herbicide.

**b)** Suggest why farmers in low-income countries may not use selective herbicides, and what they might do instead.

**S3.** Describe one change you would make to the method of the fertiliser investigation to be more sure of the results. Give a reason for your answer.

## ●●● Challenge Explain in full sentences.

**C1.** An apple breeder has a yellow apple variety that is not sweet and a red apple variety that is sweet. Describe how the breeder could produce a new yellow, sweet variety. Use the words cross-breeding, select and generations.

**C2.** Farms in the UK today produce 35% more food per square metre than they did in 1973. Explain three different ways farmers have achieved this.

## My thinking (any language)

Before a farm is allowed to expand into a wild area, a plant ecologist carries out an ecological audit. Suggest three pieces of information the ecologist should record, and explain why each one matters.

**Word bank** mineral salts · artificial · manure · decomposers · yield · fertiliser · pesticide · insecticide · fungicide · herbicide · weed · variety · selective breeding

✂ ---------- cut here

**Exit ticket 9B Lesson 5: Growing crops** Name _____

**1.** What do fertilisers contain that helps plants grow?

**2.** What do herbicides kill, and why does this help the crop?

**3.** Describe one step in selective breeding.

**Today I can:** Goal 1 ☐ Goal 2 ☐